

Networking Notes for Exam Preparation

I. Topology Basics (All Types)

- **Definition:** Physical or logical arrangement of network devices.
- **Bus Topology:**
 - Single backbone cable connects all devices.
 - Simple, inexpensive.
 - Failure of backbone cable brings down the entire network.
- **Star Topology:**
 - Central hub/switch connects all devices.
 - Easy to install and manage.
 - Failure of central device brings down the entire network.
 - Commonly used in LANs.
- **Ring Topology:**
 - Devices connected in a closed loop.
 - Data travels in one direction.
 - Failure of one device can disrupt the network (unless dual ring is used).
- **Mesh Topology:**
 - Every device connected to every other device (full mesh).
 - Highly redundant and fault-tolerant.
 - Expensive to implement due to many connections.
 - Partial mesh: Some devices connected to multiple others.
- **Tree Topology (Hierarchical):**
 - Combination of bus and star topologies.
 - Central root node, with branches connecting other nodes.
 - Scalable.
- **Hybrid Topology:**
 - Combination of two or more different topologies.

II. Types of Communication

- **Unicast:**
 - One-to-one communication.
 - Data sent from a single source to a single destination.
 - Example: Web Browse.
- **Multicast:**
 - One-to-many communication.
 - Data sent from a single source to a specific group of destinations.
 - Example: Video conferencing to a defined group.
- **Broadcast:**
 - One-to-all communication.
 - Data sent from a single source to all devices on a network segment.
 - Example: ARP requests.

III. Half Duplex & Full Duplex

- **Half Duplex:**
 - Data can flow in both directions, but not simultaneously.
 - Devices take turns sending and receiving.
 - Example: Walkie-talkie.
- **Full Duplex:**
 - Data can flow in both directions simultaneously.
 - Allows for faster communication.
 - Example: Telephone conversation.

IV. TCP, UDP, Protocol

- **Protocol:**
 - A set of rules governing data communication.
 - Defines format, timing, sequencing, and error control.
- **TCP (Transmission Control Protocol):**
 - **Connection-oriented:** Establishes a connection before data transfer.
 - **Reliable:** Ensures delivery of all data packets.
 - **Ordered:** Guarantees data arrives in the correct sequence.
 - **Error-checked:** Detects and retransmits lost or corrupted data.
 - Used for applications requiring high reliability (e.g., HTTP, FTP, Email).
- **UDP (User Datagram Protocol):**
 - **Connectionless:** Does not establish a connection before sending data.
 - **Unreliable:** Does not guarantee delivery or order of packets.
 - **Faster:** Less overhead due to lack of reliability mechanisms.
 - Used for applications where speed is more critical than reliability (e.g., DNS, VoIP, streaming video).

V. Based on Processing: Distributed, Centralized

- **Centralized Processing:**
 - All processing occurs on a single, powerful central computer.
 - Clients submit requests to the central server.

- Easier to manage and secure initially.
- Single point of failure.
- Performance bottleneck if load is high.
- **Distributed Processing:**
 - Processing tasks are spread across multiple computers in a network.
 - Increases fault tolerance and scalability.
 - More complex to design and manage.
 - No single point of failure for the entire system.

VI. Media Access Method

- **Definition:** Rules determining which device can transmit data on a shared medium at a given time.
- **MAC Address (Media Access Control Address):**
 - Unique physical address assigned to network interfaces (NICs).
 - Burned into the hardware.
 - Used at the Data Link Layer (Layer 2) for local communication.
 - Globally unique.

VII. Network Expansion Devices

- **NIC Card (Network Interface Card):**
 - Hardware component that connects a computer to a network.
 - Responsible for physical connection and data transmission/reception.
- **Repeater:**
 - Amplifies and regenerates network signals to extend network distance.
 - Operates at Physical Layer (Layer 1).
- **Hub:**
 - Central connection point for devices in a star topology.
 - Broadcasts all incoming data to all connected devices.
 - Operates at Physical Layer (Layer 1). Creates collision domains.
- **Switch:**
 - Intelligent device that forwards data only to the intended destination based on MAC addresses.
 - Operates at Data Link Layer (Layer 2).
 - Reduces collisions and improves network performance.
- **Bridge:**
 - Connects two LAN segments and filters traffic based on MAC addresses.
 - Operates at Data Link Layer (Layer 2).
- **Router:**
 - Connects different networks (e.g., LANs to WANs).
 - Forwards data packets based on IP addresses.
 - Operates at Network Layer (Layer 3).
 - Determines best path for data.
- **Gateway:**
 - Connects networks that use different protocols.
 - Translates protocols to allow communication between disparate networks.
 - Can operate at any layer of the OSI model.
- **Firewall:**
 - Network security system that controls incoming and outgoing network traffic.
 - Based on predetermined security rules.
 - Protects networks from unauthorized access.

VIII. LAN, WAN, MAN

- **LAN (Local Area Network):**
 - Connects devices within a small, limited geographical area (e.g., home, office, school).
 - High data transfer rates.
 - Owned by a single organization or individual.
- **WAN (Wide Area Network):**
 - Connects devices over a large geographical area (e.g., cities, countries, continents).
 - Uses public or leased communication lines.
 - Lower data transfer rates compared to LANs.
 - Example: The Internet.
- **MAN (Metropolitan Area Network):**
 - Spans a metropolitan area, larger than a LAN but smaller than a WAN.
 - Typically connects multiple LANs within a city.
 - Often owned by a single entity (e.g., municipality) or used for public services.

IX. Ethernet

- **Definition:** A family of computer networking technologies for LANs.
- **Thicknet (10BASE5):**
 - Original Ethernet standard.
 - Uses thick coaxial cable.
 - Difficult to install, outdated.
- **Thinnet (10BASE2):**
 - Uses thinner coaxial cable.
 - Easier to install than Thicknet, but still largely replaced.
- **Ethernet CSMA/CD (Carrier Sense Multiple Access with Collision Detection):**
 - A media access control method used in early Ethernet networks.
 - **Carrier Sense:** Devices listen before transmitting to check if the line is free.
 - **Multiple Access:** Multiple devices can access the network.

- **Collision Detection:** If two devices transmit simultaneously (collision), they detect it, stop, and retransmit after a random delay.
- Less efficient in high traffic, mostly replaced by switches in modern networks.

X. Token Ring

- **Definition:** A LAN technology where devices are connected in a ring topology.
- **Token Passing:** A special frame called a "token" circulates around the ring.
- Only the device holding the token can transmit data.
- Provides deterministic access to the network (no collisions).
- Less common now, largely replaced by Ethernet.

XI. FDDI (Fiber Distributed Data Interface)

- **Definition:** A high-speed LAN technology, primarily using fiber optic cables.
- **Dual Ring:** Uses a dual counter-rotating ring for redundancy.
- Provides high bandwidth and fault tolerance.
- More robust and faster than Token Ring, but also largely superseded by faster Ethernet.

XII. OSI Layer (Open Systems Interconnection Model)

- **Definition:** A conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers.
- **Purpose:** Helps understand how data travels through a network and allows different systems to communicate.
- **Seven Layers (from top to bottom):**
 1. **Application Layer (Layer 7):** Provides network services to end-user applications (e.g., HTTP, FTP, SMTP).
 2. **Presentation Layer (Layer 6):** Handles data format, encryption, and compression (e.g., JPEG, MPEG).
 3. **Session Layer (Layer 5):** Manages communication sessions between applications (e.g., NetBIOS, RPC).
 4. **Transport Layer (Layer 4):** Provides end-to-end communication, reliability, flow control (e.g., TCP, UDP).
 5. **Network Layer (Layer 3):** Handles logical addressing (IP addresses) and routing of packets (e.g., IP, ICMP).
 6. **Data Link Layer (Layer 2):** Provides physical addressing (MAC addresses), error detection, and flow control on a local network (e.g., Ethernet, ARP).
 7. **Physical Layer (Layer 1):** Defines physical characteristics of the network medium (cables, connectors, voltage levels) (e.g., Cables, Hubs, Repeaters).

XIII. VLAN's Lookup & Netstat

- **VLAN (Virtual Local Area Network):**
 - Logically segments a physical LAN into multiple broadcast domains.
 - Allows devices on different physical switches to be on the same logical network.
 - Improves security, efficiency, and network management.
- **Netstat (Network Statistics):**
 - Command-line utility to display network connections (incoming/outgoing), routing tables, and network interface statistics.
 - Useful for troubleshooting network issues and monitoring network activity.

XIV. Tracert & Latency

- **Traceroute (Tracert on Windows, traceroute on Linux/macOS):**
 - Command-line utility used to trace the path that data packets take from source to destination.
 - Displays the IP addresses of routers (hops) along the path and the time taken for each hop.
 - Useful for diagnosing network connectivity issues and identifying bottlenecks.
- **Latency:**
 - The time delay between the cause and effect of a physical change in the system being observed.
 - In networking, it's the time it takes for a data packet to travel from its source to its destination.
 - Measured in milliseconds (ms). High latency means slow communication.

XV. TTL (Time to Live)

- **Definition:** A mechanism that limits the lifespan or lifetime of data in a computer or computer network.
- **In IP packets:** A field in the IP header that specifies the maximum number of hops (routers) a packet can traverse before being discarded.
- **Purpose:** Prevents packets from circulating indefinitely in a network (routing loops).
- Each router decrements the TTL value; if it reaches zero, the packet is dropped.

XVI. IP Address

- **Definition:** A numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.
- **Purpose:** Identifies a device on a network and its location.
- **Types:**
 - **IPv4:** 32-bit address (e.g., 192.168.1.1).
 - **IPv6:** 128-bit address (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

XVII. Subnet Mask

- **Definition:** A 32-bit number that distinguishes the network portion of an IP address from the host portion.
- **Purpose:** Divides an IP network into smaller subnetworks (subnets).
- Allows for more efficient use of IP addresses and better network management.

XVIII. Loopback Address

- **Definition:** A special IP address (127.0.0.1 for IPv4, ::1 for IPv6) that refers to the local machine itself.
- **Purpose:** Used for testing network applications and services locally without needing a physical network connection.
- Data sent to the loopback address never leaves the device.

XIX. Broadcast Address

- **Definition:** An IP address used to send data to all devices on a specific network or subnet.
- **Purpose:** Allows a single message to reach all hosts on a particular network segment.
- In IPv4, the broadcast address is the highest IP address in a given network range (e.g., for 192.168.1.0/24, the broadcast address is 192.168.1.255).

XX. Continuous Ping Test

- **Ping (Packet Internet Groper):**
 - Network utility used to test the reachability of a host on an Internet Protocol (IP) network.
 - Measures the round-trip time for messages sent from the originating host to a destination computer and back.
- **Continuous Ping:**
 - Running the ping command continuously (e.g., `ping -t <IP address>` on Windows, `ping <IP address>` on Linux/macOS).
 - **Purpose:**
 - Monitor network connectivity over time.
 - Detect intermittent network issues or packet loss.
 - Assess network stability and latency.